



Class 9 2022-23

ATOMS AND MOLECULES

Ch 3 | Class 9th (Science)

Handwritten Notes & Imp Questions



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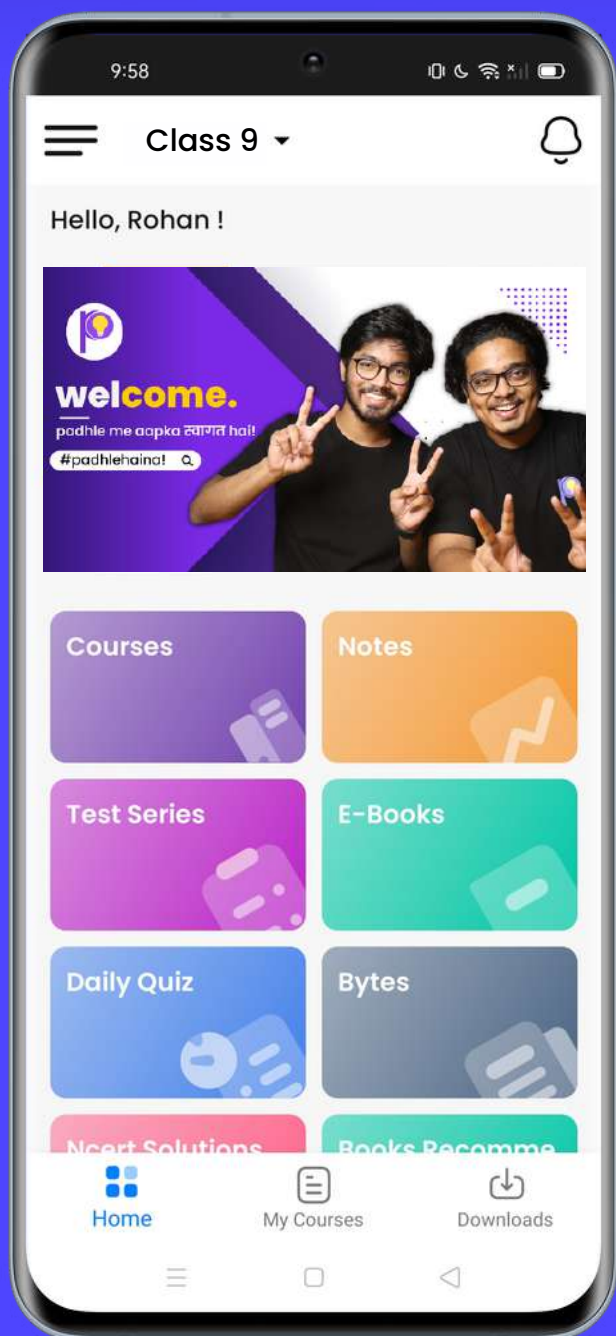
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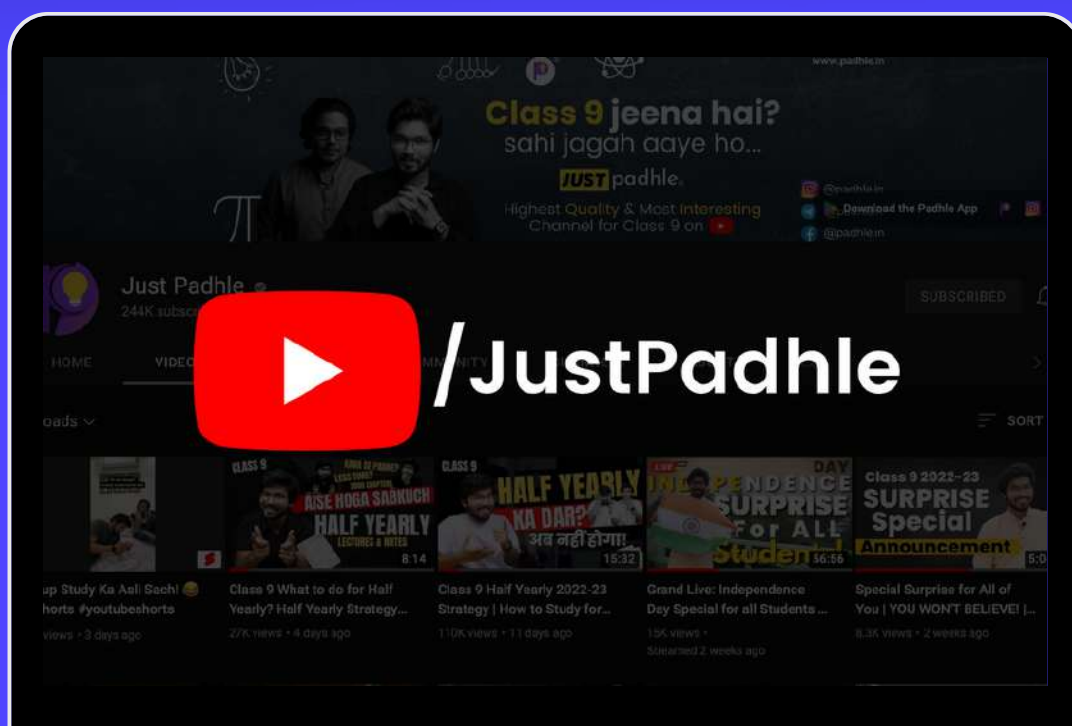
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Atoms And Molecules

(आखिर ये संसार बना किससे है ?)

Let's start with some definitions —

• Atoms :

The smallest entity of an element, which shows the properties of that element.

• Molecules :

The smallest entity of a compound.

- Now, all these reactions and combinations were going on, so we needed to find some laws that can govern the chemical reactions.

→ Gattu loves football, but when he doesn't know the laws of football then how will he play it?

Laws Of Chemical Combination:

- This law was established after the experiments by Lavoisier and Joseph L. Proust.
- The chemical reaction between two or more substances gives rise to products which is governed by certain laws called Laws of Chemical Combination.

LAW OF CONSERVATION OF MASS

- During a chemical reaction, the total mass of reactants will be equal to the total mass of the products.
- Mass can neither be created nor destroyed in a chemical reaction.
- Example: $A(\text{reactant}) + B(\text{reactant}) \longrightarrow AB(\text{product})$
mass of A + mass of B = mass of AB.

LAW OF CONSTANT PROPORTIONS:

◇ In a chemical reaction, compounds always contain the same elements present in definite proportions by mass irrespective of their source.

◇ It was given by Lavoisier.

◇ For example:

(i) 18 gm of H_2O = 2 gm of hydrogen + 16 gm of oxygen

⇒ Mass of hydrogen/mass of oxygen = $2/16 = 1/8$

ii) 36 gm of H_2O = 4 gm of hydrogen + 32 gm of oxygen

⇒ Mass of hydrogen/mass of oxygen = $4/32 = 1/8$

iii) 9 gm of H_2O = 1 gm of hydrogen + 8 gm of oxygen

⇒ mass of hydrogen/mass of oxygen = $1/8$

• This verifies law of constant proportions as the ratio of mass of hydrogen to oxygen is always same.

DALTON'S ATOMIC THEORY

- Every matter is made up of very small or tiny particles called atoms.
- Atoms are not divisible and cannot be created or destroyed in a chemical reaction.
- All atoms of a given element are same in size, mass and chemical properties.
- Atoms of different elements are different in size, mass and chemical properties.
- Atoms combine in the ratio of a small whole number to form compounds.
- The relative number and kinds of atoms are constant in a given compound.

* Atoms की विस्तृत चर्चा :

ATOMS

- Smallest unit of an element.
- May or may not exist independently, but takes part in chemical reaction.
- Radius of an atom is Atomic Radius, generally in nanometers.

Symbol of atoms

✧ The first letter of a symbol is always written as a capital letter (uppercase) and the second letter as a small letter (lowercase).
For example: Hydrogen (H), aluminium (Al), cobalt (Co).

→ Some other symbols have been taken from the names of elements in Latin, German or Greek.

For eg.: Fe from its Latin name ferrum, sodium is Na from natrium, potassium is K from kalium.

Element	Symbol	Element	Symbol	Element	Symbol
Aluminium	Al	Copper	Cu	Nitrogen	N
Argon	Ar	Fluorine	F	Oxygen	O
Barium	Ba	Gold	Au	Potassium	K
Boron	B	Hydrogen	H	Silicon	Si
Bromine	Br	Iodine	I	Silver	Ag
Calcium	Ca	Iron	Fe	Sodium	Na
Carbon	C	Lead	Pb	Sulphur	S
Chlorine	Cl	Magnesium	Mg	Uranium	U
Cobalt	Co	Neon	Ne	Zinc	Zn

Atomic Mass

→ The mass of an atom of an element is called its atomic mass.

→ The atomic mass unit is defined as the quantity of mass equal to $1/12$ of mass of an atom of carbon-12.

$$1 \text{ amu or } u = \frac{1}{12} \times \text{Mass of an atom of } C^{12}.$$

$$1 u = 1.66 \times 10^{-27} \text{ kg}.$$

* ATOMIC MASS OF SOME IMPORTANT ELEMENTS (usually asked in exams)

Element	Atomic Mass (u)
Hydrogen	1
Carbon	12
Nitrogen	14
Oxygen	16
Sodium	23
Magnesium	24
Sulphur	32
Chlorine	35.5
Calcium	40

* Molecules की विस्तृत चर्चा

- A molecule is in general a group of two or more atoms that are chemically bonded together.
- A molecule is the smallest particle of matter (except element) which is capable of an independent existence and shows all properties of that substance.
- E.g.: ' H_2O ' is the smallest particle of water which shows all the properties of water.

* Molecules of Elements :

- कुछ elements इतने introvert होते हैं कि वो अकेले exist नहीं कर सकते.
- E.g. O is introvert, इसलिए वो अकेले बाहर नहीं निकलता. एक और O को ढूँढ़ के उसके साथ bond बनाके O_2 के form में exist करता है.

MOLECULE OF ELEMENTS

- Molecules of an element are constituted by the same type of atoms.
- Number of atoms constituting a molecule is called its atomicity.
- Molecules of some elements contain only one atom (monoatomic).
- Molecules of some elements contain only two atoms (diatomic).

Mono atomic	Diatomic	Triatomic	Polyatomic
Helium He,	Hydrogen H ₂ ,	Ozone O ₃	Phosphorus P ₄ ,
Copper Cu,	Oxygen O ₂		Sulphur S ₈
Krypton Kr,	Nitrogen N ₂		
Silver Ag,			
Carbon C			

Molecules of Compounds

- Atoms of different elements join together in definite proportions to form molecules of compounds.
- For eg. a molecule of water consists of two atoms of hydrogen and one atom of oxygen to form a triatomic molecule H₂O.

Ions

- Compounds composed of metals and non-metals contain charged species.
 - The charged species are known as ions.
- An ion is a charged particle and can be negatively or positively charged. (मेजा देजा)
- A negatively charged ion is called an 'anion' and the positively charged ion, a 'cation'.
- Ions can consist of a single charged atom or a group of atoms that has a net charge on them — Called Polyatomic Ion.

(i) Cation : It is positively charged ion and is formed by the loss of one or more electrons from an atom.

* For example : Sodium atom, loses one electron to form a sodium ion Na



(ii) Anion : It is a negatively charged ion and is formed by the gain of one or more electrons by an atom.

* For eg. : a chlorine atom gains one electron to form a chlorine ion Cl^- .



VALENCY

- The combining capacity of an element is known as its **Valency**.
- Valency is used to find out how the atom of an element will combine with the atom of another element to form a chemical compound.

(Every atom wants to become stable, to do so it may lose, gain or share electrons.)

- If an atom consists of 1, 2 or 3 electrons in its valence shell then its valency is 1, 2 or 3 respectively,
- If an atom consists of 5, 6 or 7 electrons in the outermost shell, then it will gain 3, 2 or 1 electron respectively and its valency will be 3, 2 or 1 respectively.
- If an atom has 4 electrons in the outermost shell then it will share this electron and hence its valency will be 4.

- If an atom has 8 electrons in the outermost shell and hence its valency will be 0.

* Valency in simple language (only for understanding)

◇ Every atom wants to have 8 electrons in its outermost shell to be stable (except H and He which only have single shell, they require only 2 electrons to be stable)

कितने electrons की हरा फेरी करनी पड़ेगी किसी atom को stable करने के लिए, और कौनसी हरा फेरी सस्ती होगी, that is valency.

For example, Na has total of 11 electrons, i.e. 1 electron in the outermost shell.

Now either you can add 7 electrons to it, or you can remove 1 electron. In both the cases the outermost shell will have 8 electrons.

But since 1 electron remove करना सस्ता है, इसलिए valency is 1.

On the other hand, Cl has 7 electrons in the outermost shell.

So either we add 1 electron to it or remove 7 electrons.

कम सेहजत किसमें लगेगी?

Ans: 1 electron remove करने में ! i.e. its valency is also 1.

सस्ती हरा फेरी is valency.

* How to write chemical formulas?

Rules for writing chemical formulae:

1. We first write symbols of elements which form compound.

2. Below the symbol of each element, we should write their valency.

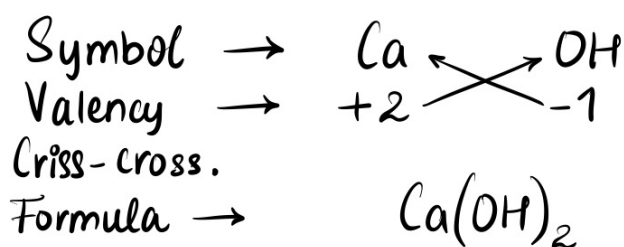
3. Now cross over the valencies of combining atoms.

4. With the first atom, we write the valency of the second atom (as a subscript).

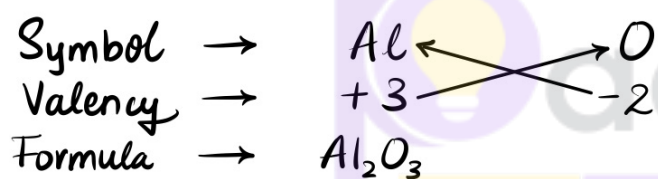
5. With the second atom, we write the valency of first atom (subscript).

Chemical formula of some simple compounds :

a) Calcium hydroxide



b) Aluminium oxide



Molecular Mass

$\rightarrow$ It is the sum of atomic masses of all the atoms in a molecule of that substance.

Example : Molecular mass of

$\text{H}_2\text{O} = 2 \times \text{Atomic mass of Hydrogen} + 1 \times \text{Atomic mass of Oxygen}$

So, Molecular mass of $\text{H}_2\text{O} = 2 \times 1 + 1 \times 16 = 18 \text{ u}$

Formula Unit Mass :

$\diamond$ It is the sum of atomic mass of ions and atoms present in formula for a compound.

Examples : In NaCl,

$$Na = 23 \text{ a.m.u.}$$

$$Cl = 35 \text{ a.m.u.}$$

$$\text{So, Formula unit mass} = 1 \times 23 + 1 \times 35.5 = 58.5 \text{ u}$$

Mole Concept (easiest explanation in Triumph Series Video)

* Simple terms जो बचपन से सुने हैं —

$$1 \text{ Dozen Bananas} = 12 \text{ Bananas}$$

$$1 \text{ Pair Bananas} = 2 \text{ Bananas}$$

As we have learned that dozen always means 12, pair always means 2, in the same way —

$$1 \text{ Mole Bananas} = 6.022 \times 10^{23} \text{ bananas}$$

$$1 \text{ Mole of Binod} = 6.022 \times 10^{23} \text{ Binods}$$

$$\text{Similarly, } 1 \text{ Mole Atoms} = 6.022 \times 10^{23} \text{ Atoms}$$

This number (6.022×10^{23}) is also called Avogadro Number (N_A)

MOLAR MASS

→ The molar mass of a substance is the mass of 1 mole of that substance.

→ It is equal to the 6.022×10^{23} atoms of that element/substance.

Examples :

a) Atomic mass of hydrogen (H) is 1 u.
Its molar mass is 1 g/mol.

b) Atomic mass of nitrogen is 14 u.
So, molar mass of nitrogen is 14 g/mol.

c) Molar mass of S_8 = Mass of S $\times$ 8 = $32 \times 8 = 256 \text{ g/mol}$.

d) Molar mass of HCl = Mass of H + Mass of Cl = $1 + 35.5 = 36.5 \text{ g/mol}$.

Important Formulae :

i) Number of moles (n) = $\frac{\text{Given mass}}{\text{Molar mass}} = \frac{m}{M}$

ii) Number of moles (n) = $\frac{\text{Given number of particles}}{\text{Avogadro's number}}$

$$n = \frac{N}{N_0}$$

iii) $\frac{m}{M} = \frac{N}{N_0}$

$$m = \frac{M \times N}{N_0}$$

iv) Percentage of any atom in given compound = $\frac{\text{Mass of element} \times 100}{\text{Mass of compound}}$

